HOME LAB PROJECT REPORT

Active Directory Home Lab

Building and securing a small Windows domain — Windows Server 2022 domain controller and a Windows 11 workstation, with departments, a protected file share, Group Policy and PowerShell automation.

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DATES
12 – 17 October 2026

PLATFORM
VMware Workstation · Windows Server 2022 · Windows 11

DOMAIN
LAB.local

1. Lab at a glance

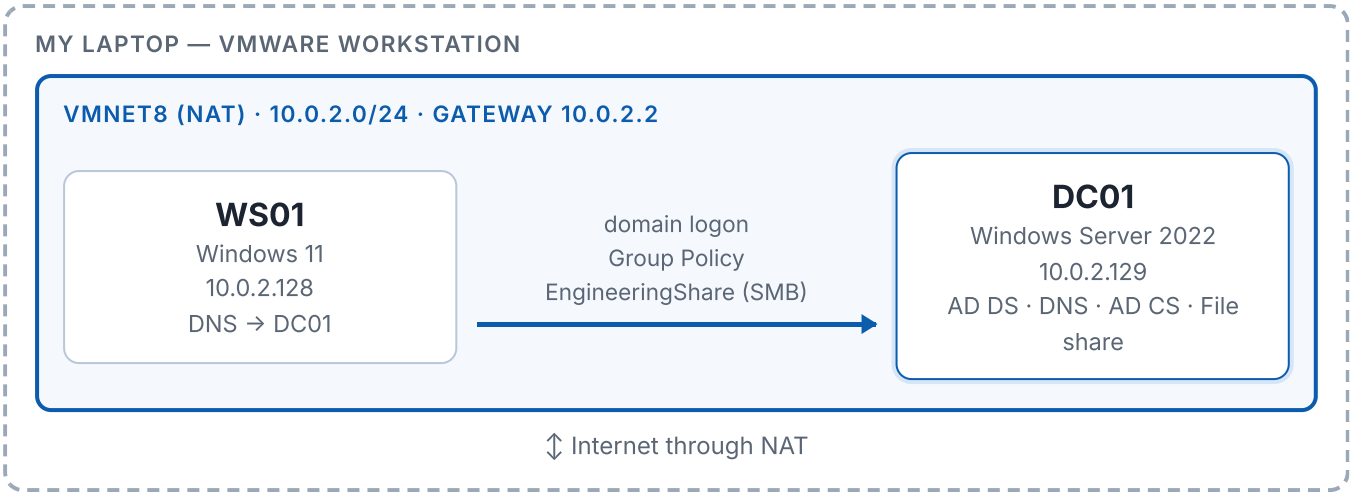
Item	Value
Domain name	`LAB.local` (short NetBIOS name: `LAB`)
Domain Controller	**DC01** — Windows Server 2022 Standard (evaluation), 8 GB RAM, 4 CPUs, 60 GB disk
Server roles	Active Directory Domain Services (AD DS), DNS Server, Active Directory Certificate Services (AD CS), File Server
Workstation	**WS01** — Windows 11, 8 GB RAM, 4 CPUs, 64 GB disk
Virtual network	VMware **VMnet8 (NAT)**, `10.0.2.0/24`, gateway `10.0.2.2`
IP addresses (fixed)	DC01 = `10.0.2.129` · WS01 = `10.0.2.128`
DNS	DC01 points to itself (`127.0.0.1`); WS01 points to DC01 (`10.0.2.129`)
Forest / domain level	Windows Server 2016
Certificate Authority	Enterprise Root CA `LAB-DC01-CA`, RSA 2048-bit key, SHA-256, valid 5 years

Directory layout I created

```
LAB.local
├── Engineering (OU)        ← Group Policy "SetEngineeringWallpaper" linked here
│   ├── john doe            (jdoe)
│   ├── lorem ispum         (lispum)
│   └── EngineeringShare    (security group)
├── Management (OU)
│   ├── Ahmed Rizwan        (ahrizwan)
│   └── hasan ali
├── IT (OU)
│   ├── Administrator
│   └── Admin (OU)
└── Computers
    └── WS01
```

EngineeringShare group members: john doe, lorem ispum, hasan ali *(hasan ali sits in Management but was added on purpose — this proves that access comes from the group, not from the department folder he lives in.)*

2. Network diagram



3. Skills shown

- Installing Windows Server 2022 in VMware and taking **snapshots** (restore points) before risky changes
- Promoting a server to a **Domain Controller** and creating a new forest (`LAB.local`)
- Setting up a **Certificate Authority** (AD CS) for the domain
- Planning a virtual network, fixed IP addresses and **DNS** so a client can find the domain
- Joining a Windows 11 computer to the domain and logging in with domain accounts
- Organising users into **Organisational Units (OUs)** by department
- Using **security groups** and **NTFS permissions** to control who can open a file share (least-access design)
- Creating and linking **Group Policy Objects (GPOs)** to a department
- Writing a **PowerShell** script to create users with random temporary passwords
- Testing every change with both an allowed user and a denied user

4. Step-by-step build

Part A — Build the server (12 Oct)

A1. Download Windows Server 2022 from the Microsoft Evaluation Center and create a new VMware virtual machine.

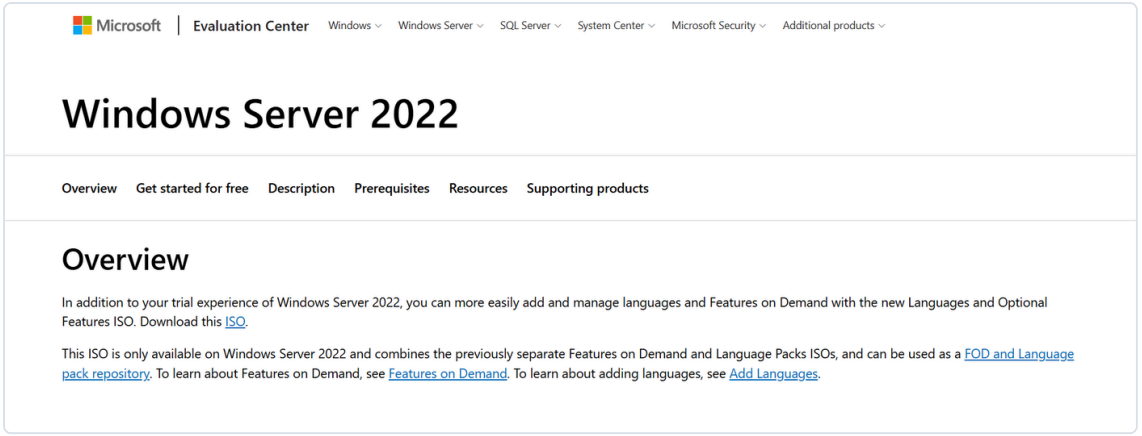


Figure 1 — Windows Server 2022 evaluation download

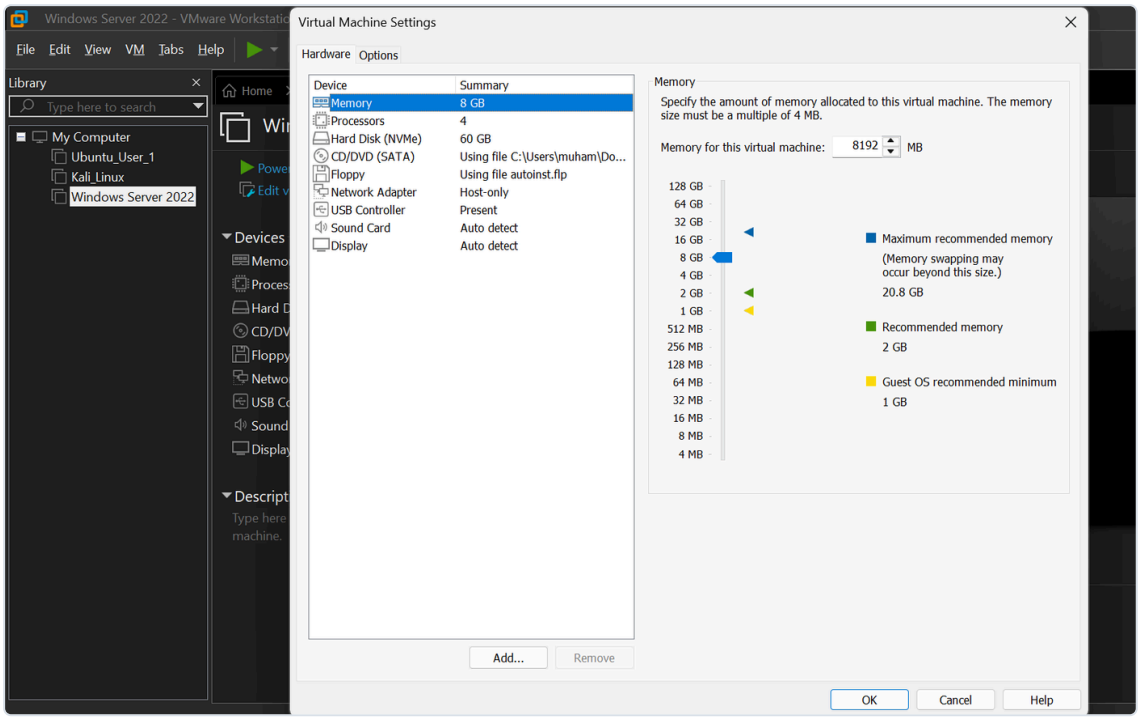


Figure 2 — Server VM hardware: 8 GB RAM, 4 processors, 60 GB disk

A2. Take a snapshot right after installing, so I can roll back if anything breaks later.

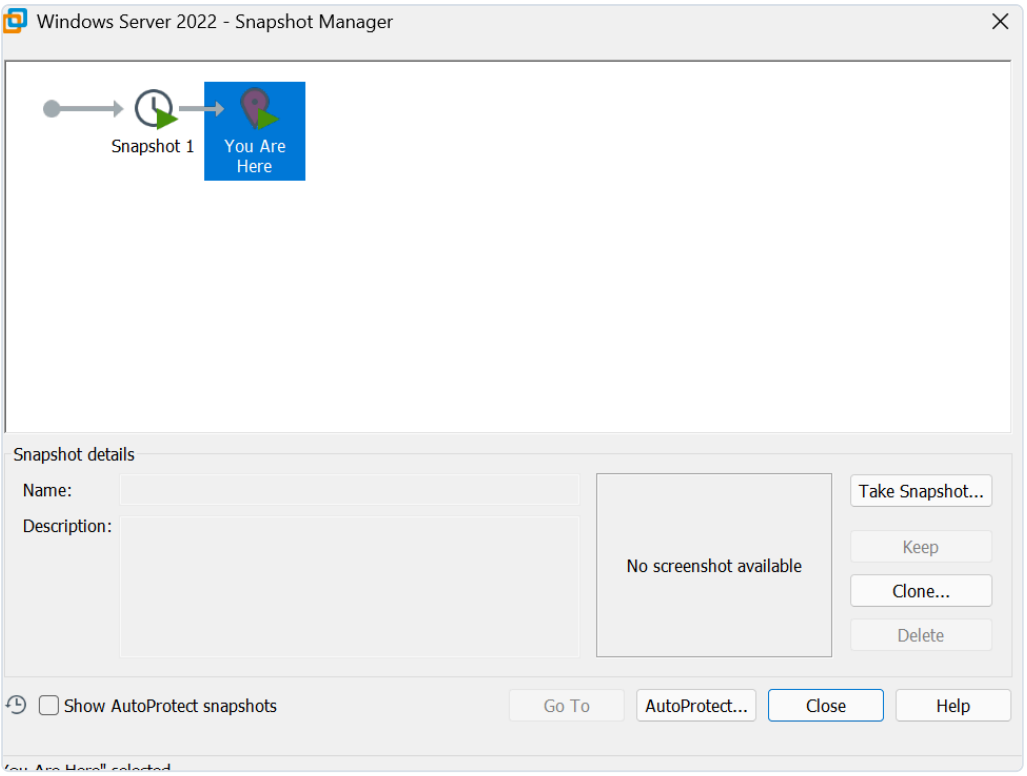


Figure 3 — "Snapshot 1" saved as a clean restore point

A3. Rename the server to `DC01` (a clear name before it becomes a domain controller, because renaming afterwards is harder).

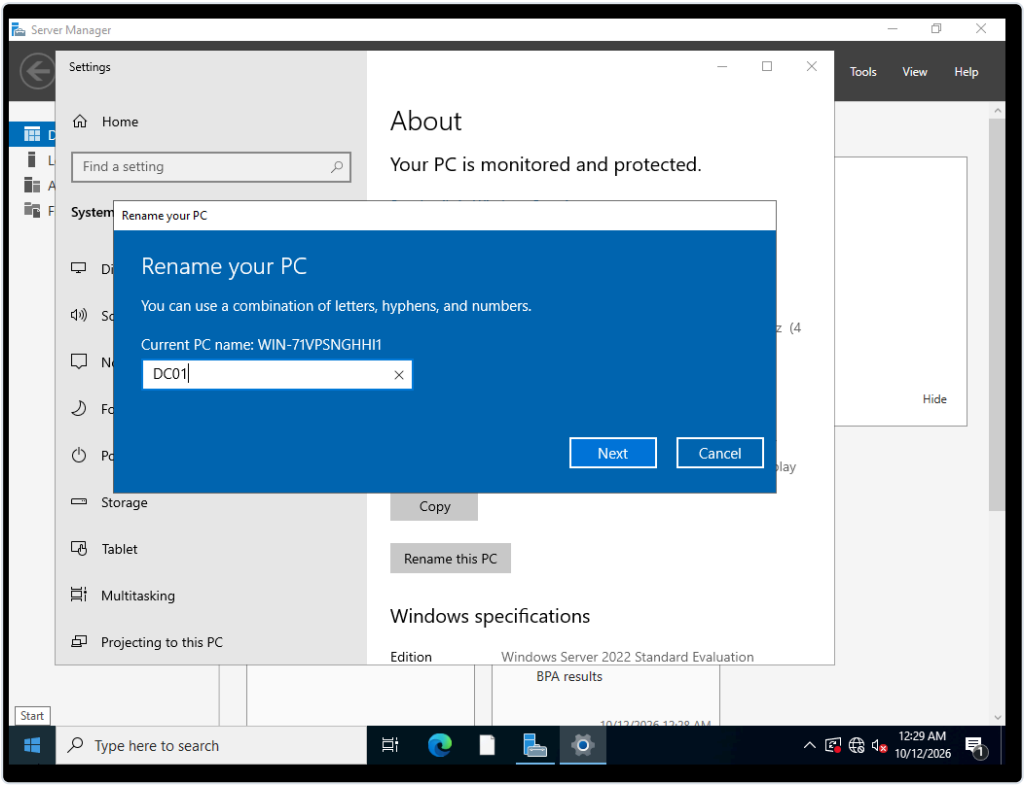


Figure 4 — Renaming the server to DC01

A4. Install Active Directory Domain Services through *Server Manager → Manage → Add Roles and Features*.

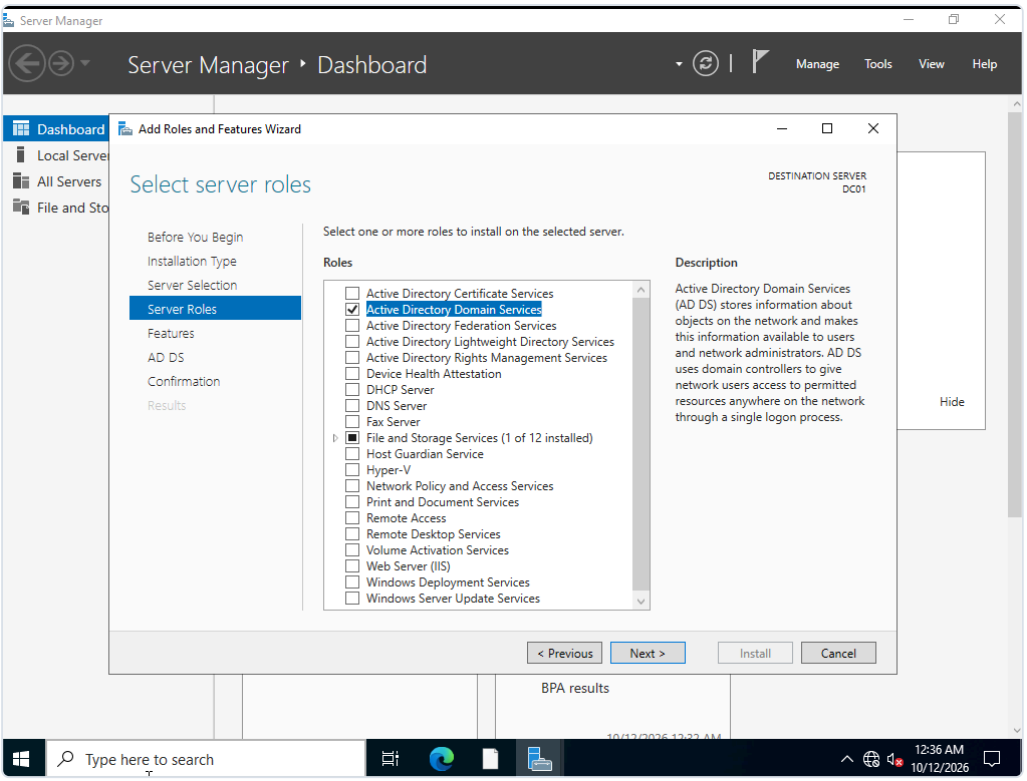


Figure 5 — Selecting the Active Directory Domain Services role

A5. Promote the server to a Domain Controller and create a new forest called `LAB.local`. I kept DNS Server and Global Catalog ticked and set a recovery (DSRM) password.

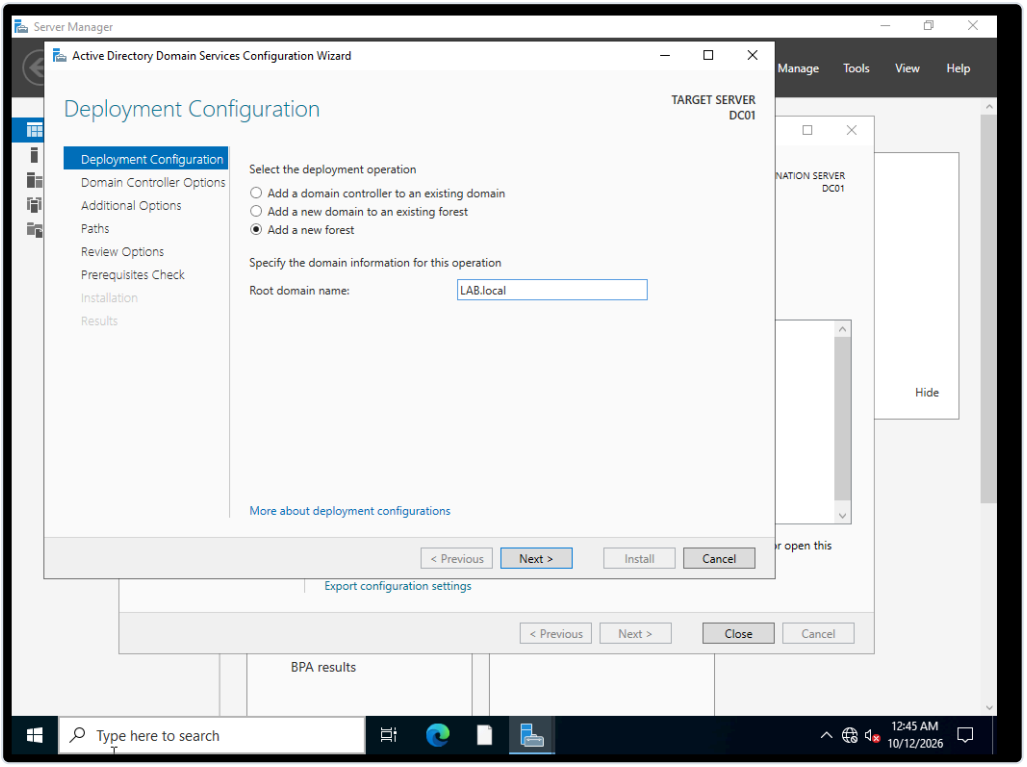


Figure 6 — New forest, root domain LAB.local

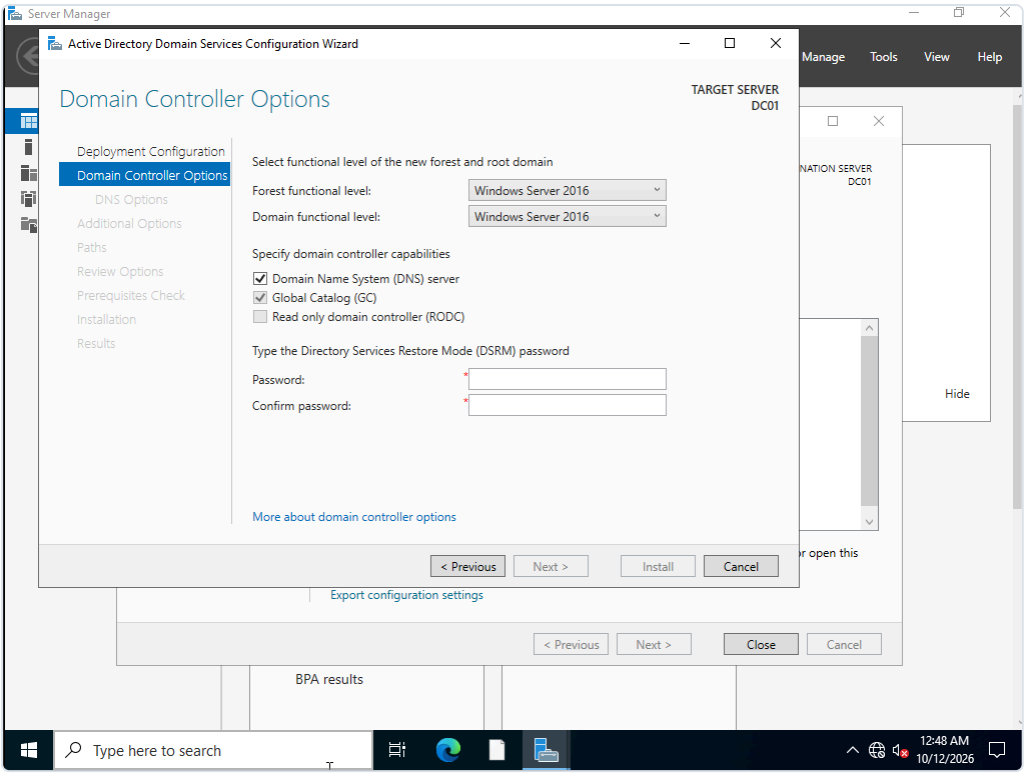


Figure 7 — Functional level Windows Server 2016, DNS and Global Catalog enabled

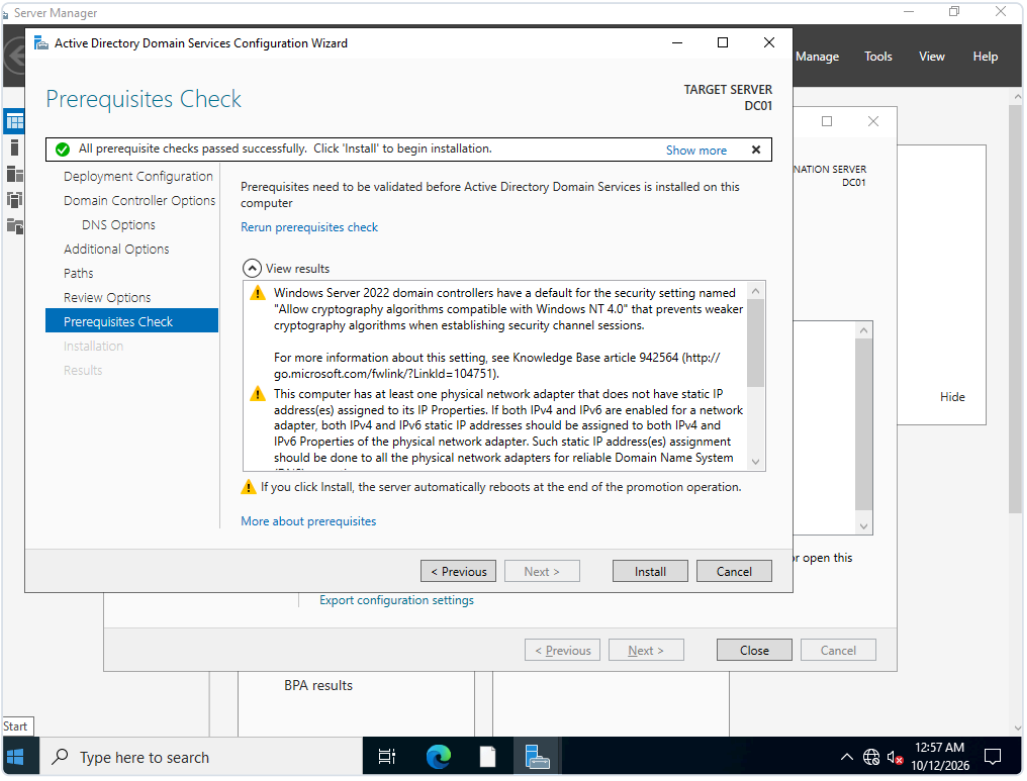


Figure 8 — Prerequisite check passed. One warning says the server has no fixed IP address yet — I fix this in Part C.

A6. Install Active Directory Certificate Services and configure an **Enterprise Root Certificate Authority**. This lets the domain issue its own certificates, which secure versions of protocols (for example LDAPS, encrypted directory lookups) need.

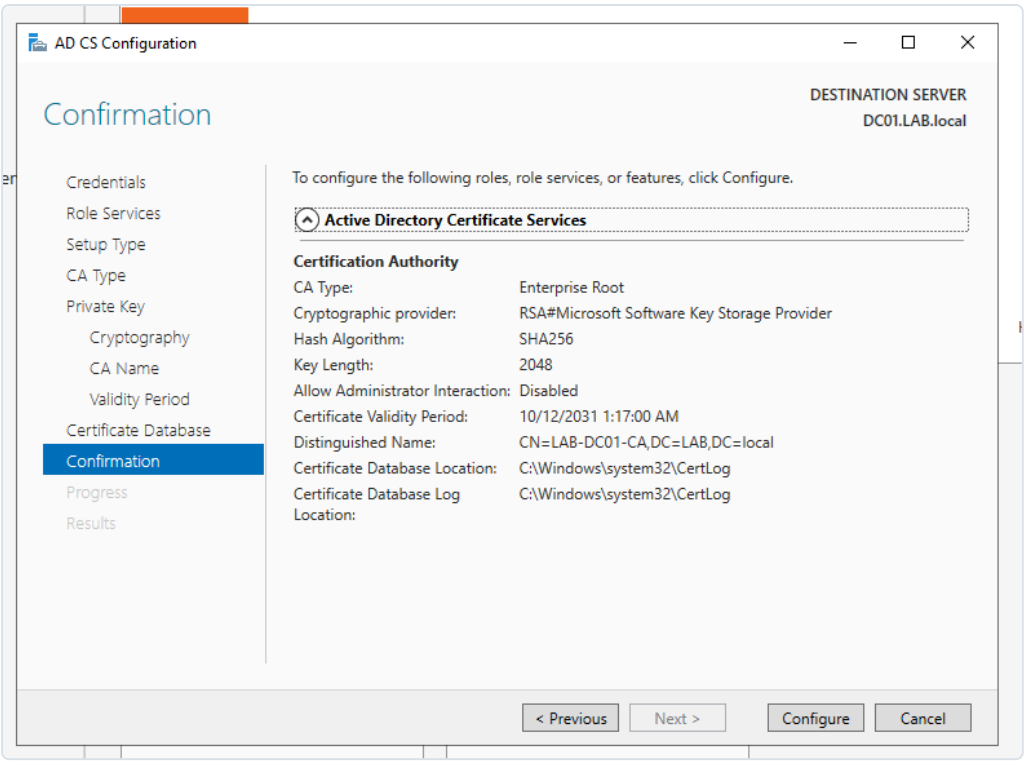


Figure 9 — CA type Enterprise Root, name LAB-DC01-CA, 2048-bit key, SHA-256

Part B — Create domain users (16 Oct)

Opened *Active Directory Users and Computers* and created four users: Ahmed Rizwan, hasan ali, john doe and lorem ispum.

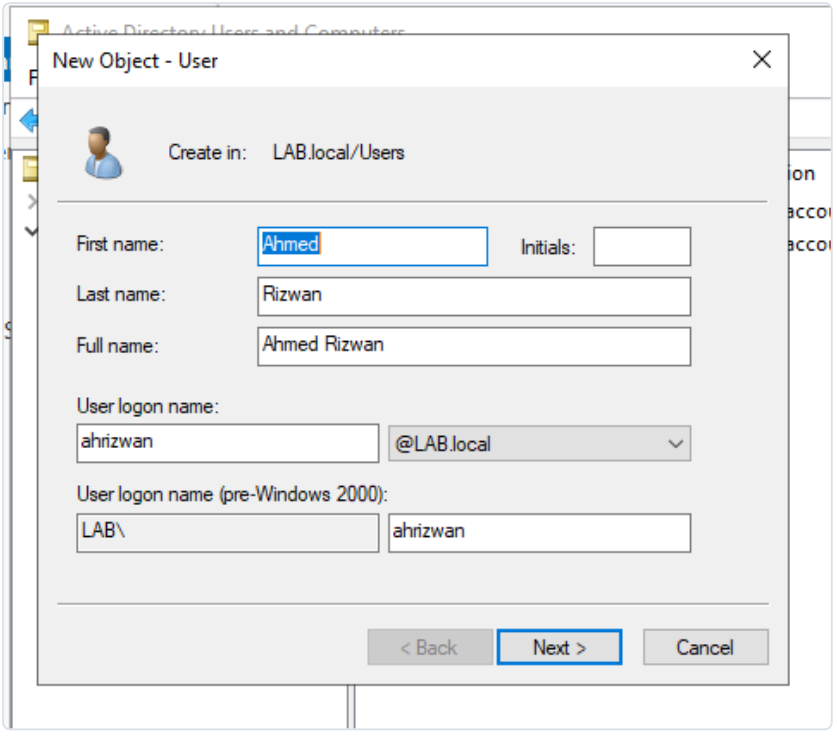


Figure 10 — Creating Ahmed Rizwan, logon name ahrizwan@LAB.local

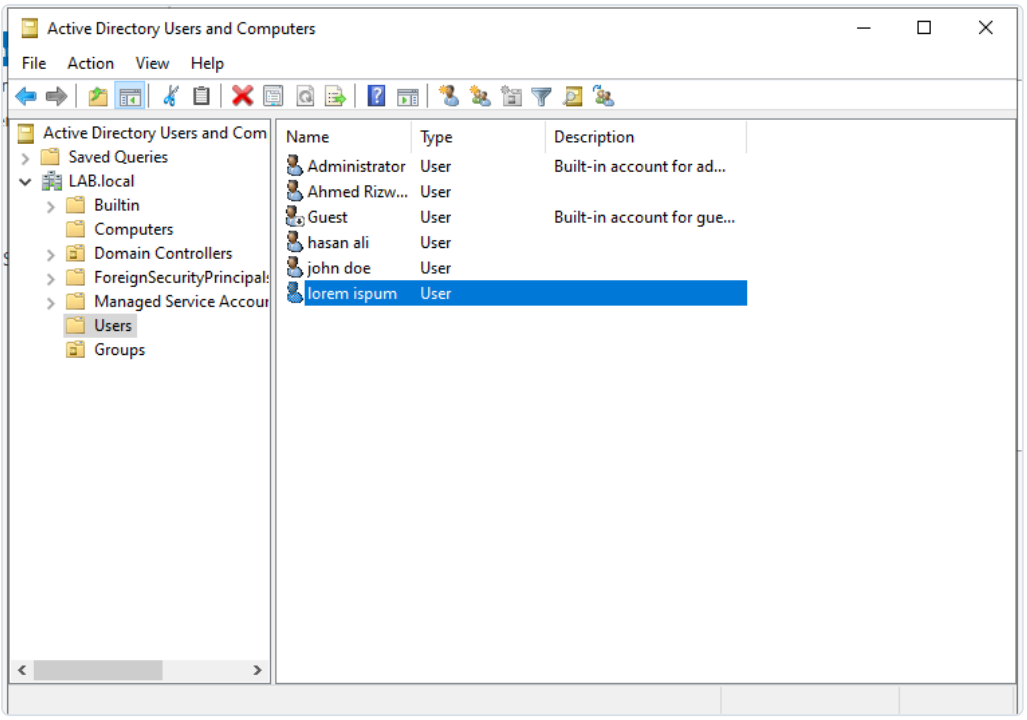


Figure 11 — All four users created

Part C — Network and fixed IP addresses (16 Oct)

C1. The server was first on VMware's *host-only* network with an automatic address. I moved **both VMs to VMnet8 (NAT)** and set its subnet to `10.0.2.0/24`, so the server and the workstation share one network that can also reach the internet.

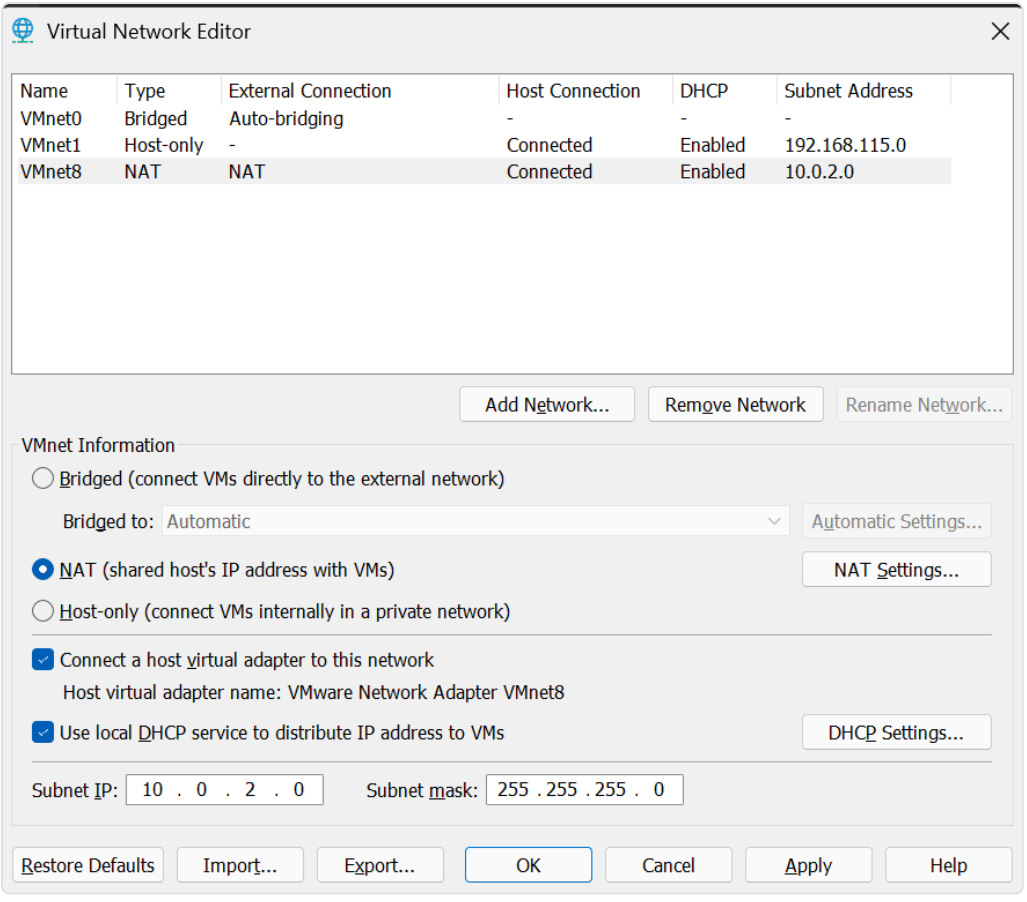


Figure 12 — VMnet8 set to NAT, subnet 10.0.2.0

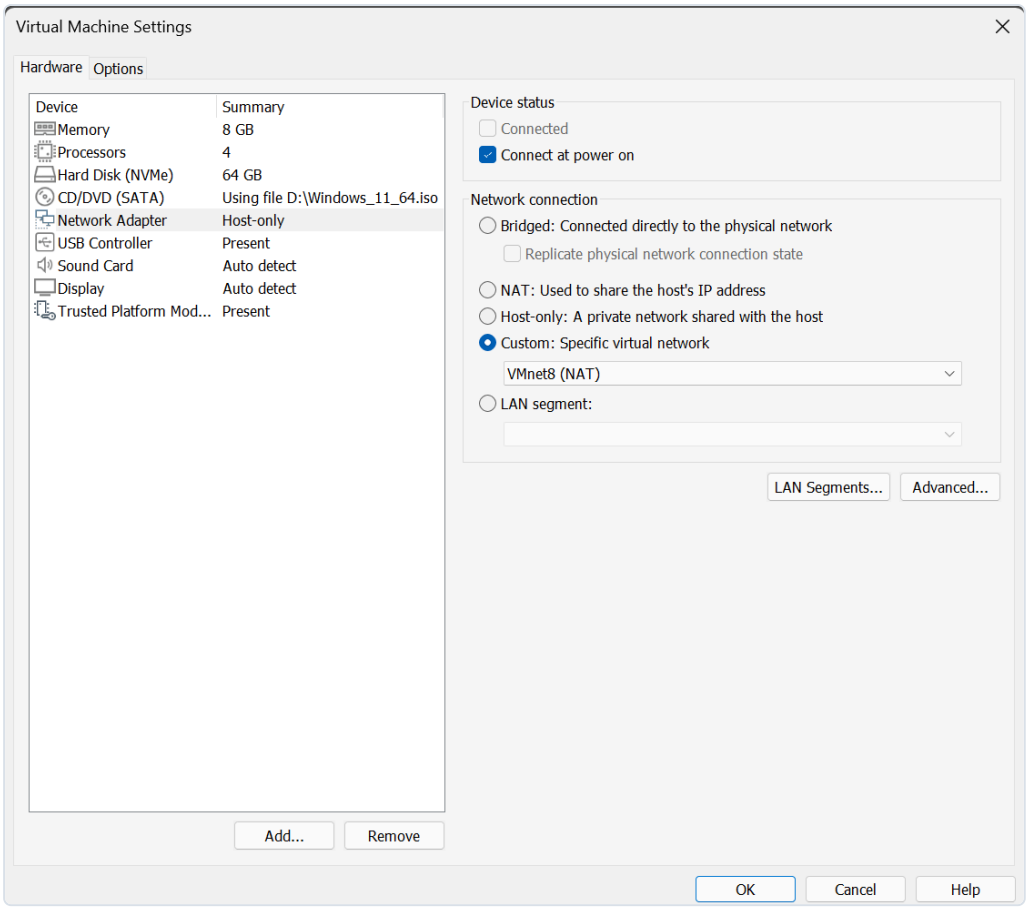


Figure 13 — Windows 11 VM network adapter set to VMnet8 (NAT)

C2. Give DC01 a fixed IP address. A domain controller must never change address, because every computer finds the domain through it. DNS is set to `127.0.0.1` (itself), because DC01 is the domain's DNS server.

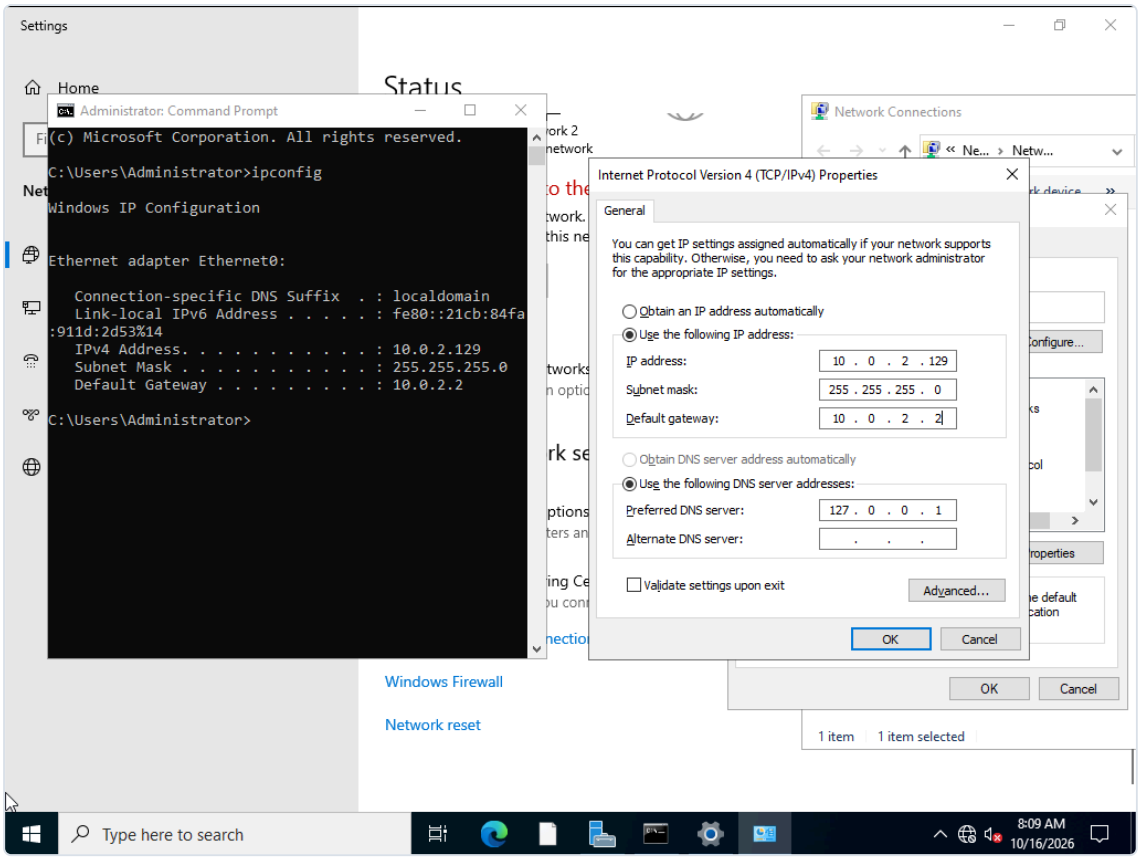


Figure 14 — DC01: IP 10.0.2.129, gateway 10.0.2.2, DNS 127.0.0.1

Part D — Join the Windows 11 workstation (16 Oct)

D1. Rename the workstation to `WS01`.

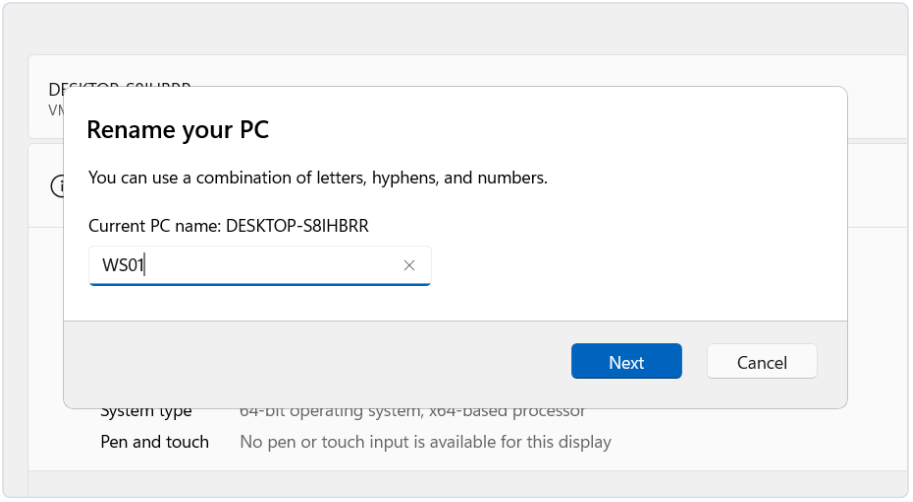


Figure 15 — Renaming the client to WS01

D2. Fixed IP and DNS pointing to DC01. This is the key step: a computer can only find the domain `LAB.local` if it asks the domain's own DNS server.

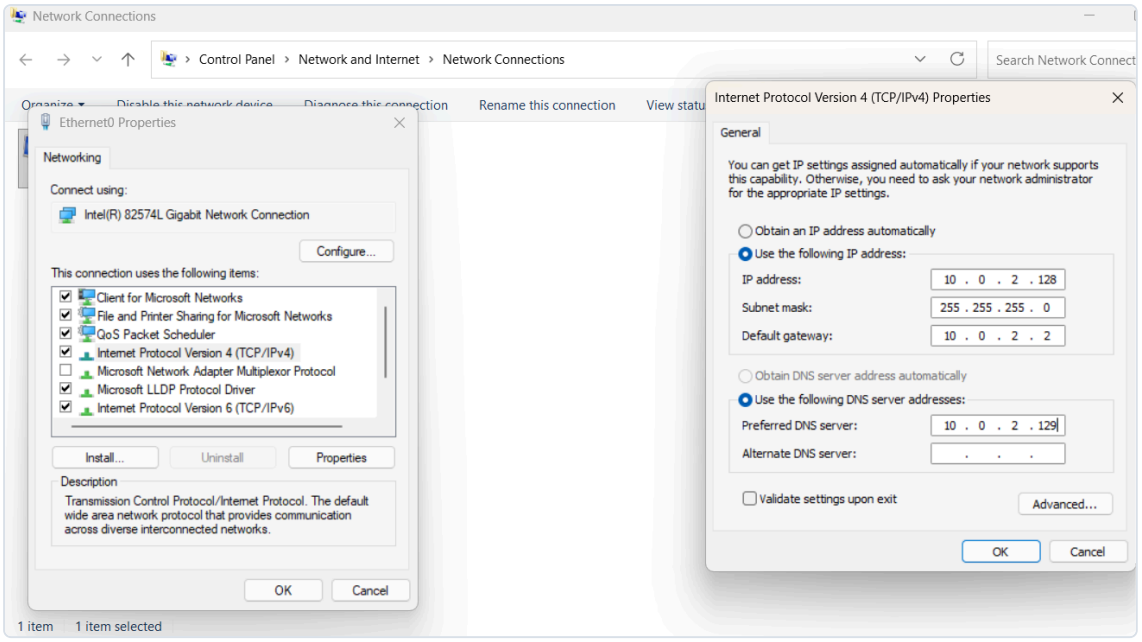


Figure 16 — WS01: IP 10.0.2.128, gateway 10.0.2.2, DNS 10.0.2.129 (DC01)

D3. Join the domain from *Settings → Accounts → Access work or school → Join this device to a local Active Directory domain*.

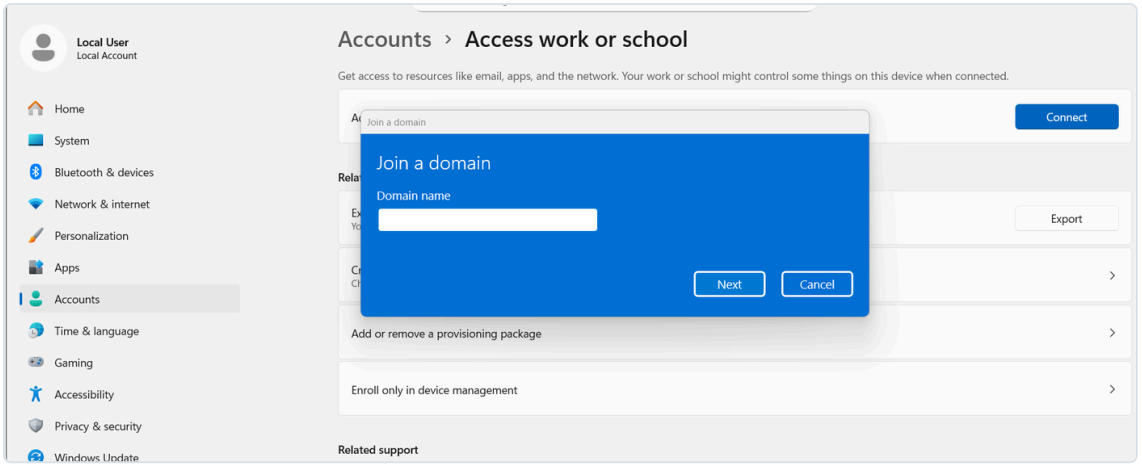


Figure 17 — Joining WS01 to LAB.local

D4. Log in as a domain user and confirm WS01 appears on the server.

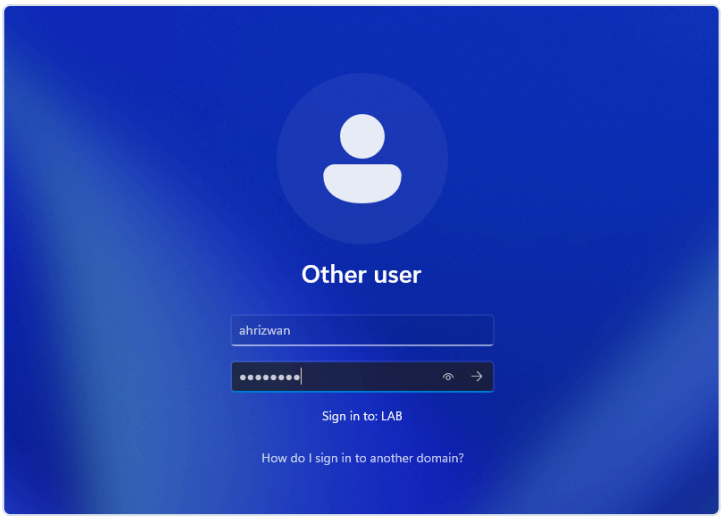


Figure 18 — Logging in to WS01 as ahrizwan ("Sign in to: LAB")

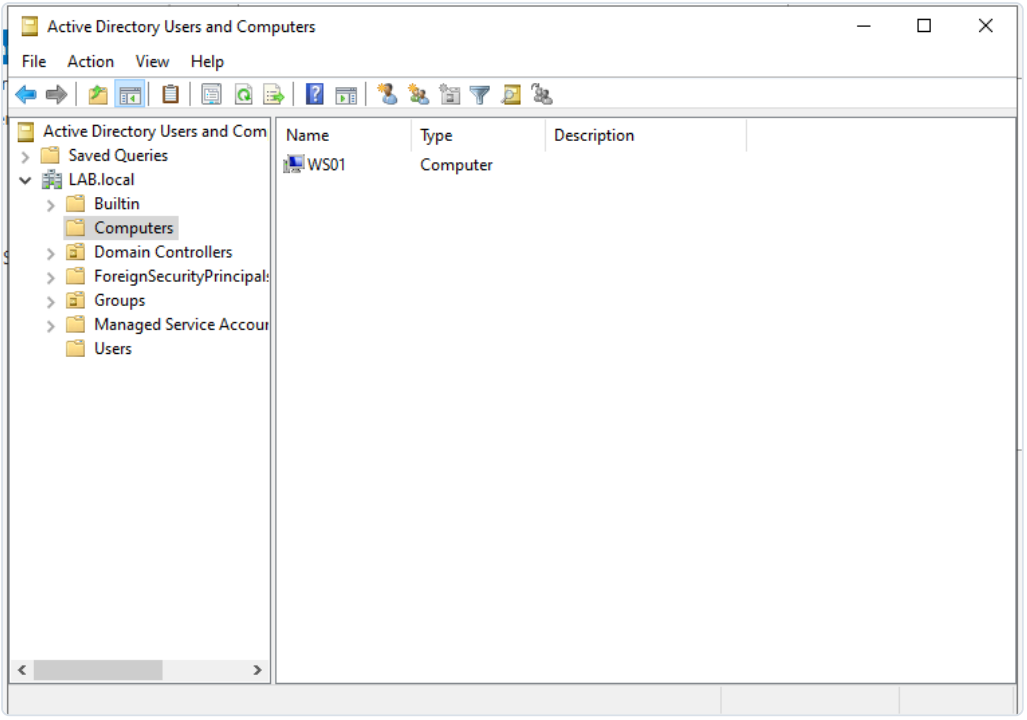


Figure 19 — WS01 now listed under Computers on DC01

Part E — Organise users into departments (17 Oct)

Created **Organisational Units** — Engineering, Management and IT (with an Admin OU inside IT) — and moved each user into their department. This matters because Group Policy can be linked to an OU, but **not** to the built-in "Users" folder.

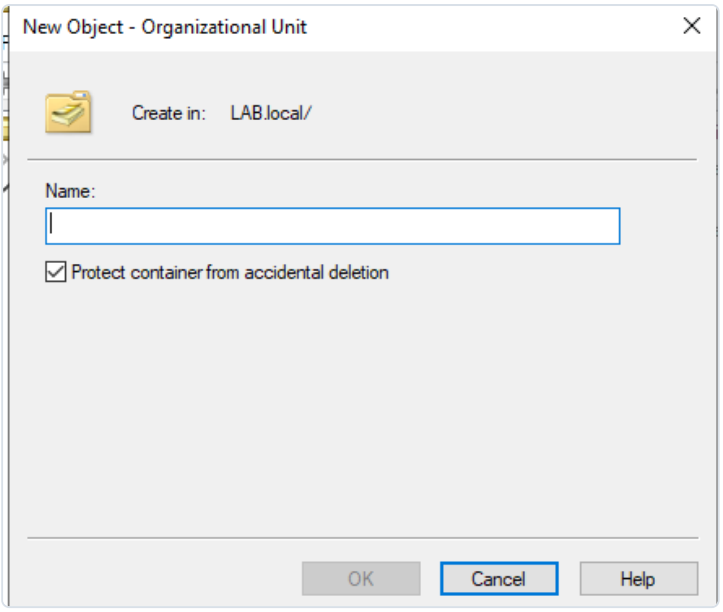


Figure 20 — Creating an OU, with "Protect from accidental deletion" ticked

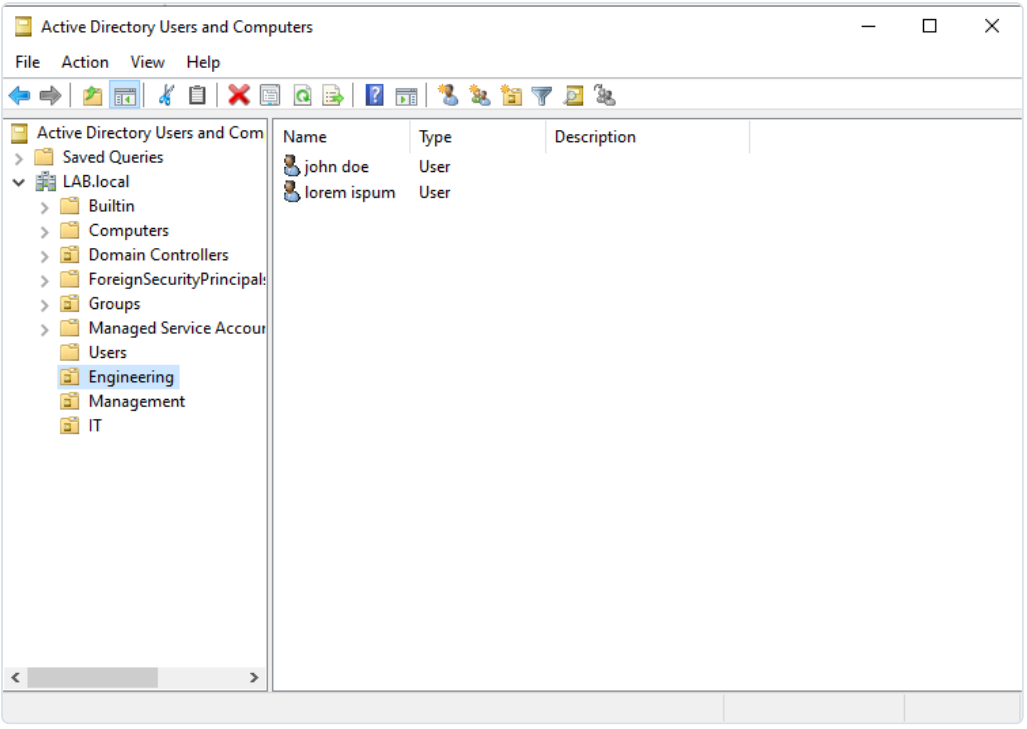


Figure 21 — Engineering: john doe, lorem ispum

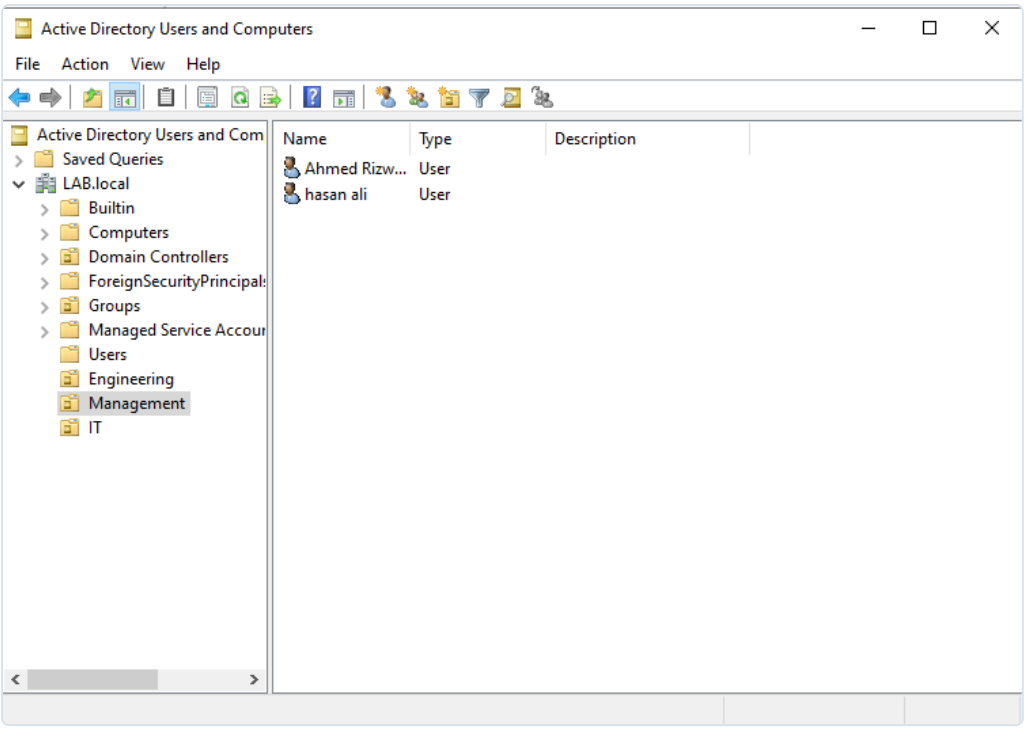


Figure 22 — Management: Ahmed Rizwan, hasan ali

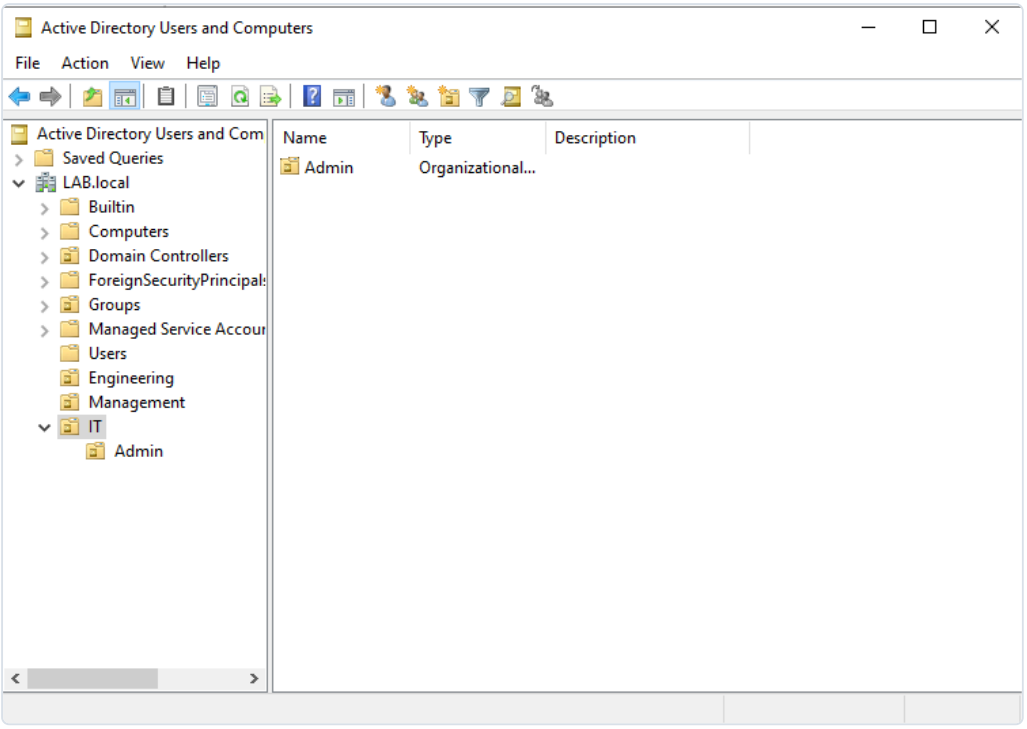


Figure 23 — IT, with a nested Admin OU

Part F — Security group (17 Oct)

Created a **Global Security group** called `EngineeringShare` and added john doe, lorem ispum and hasan ali.

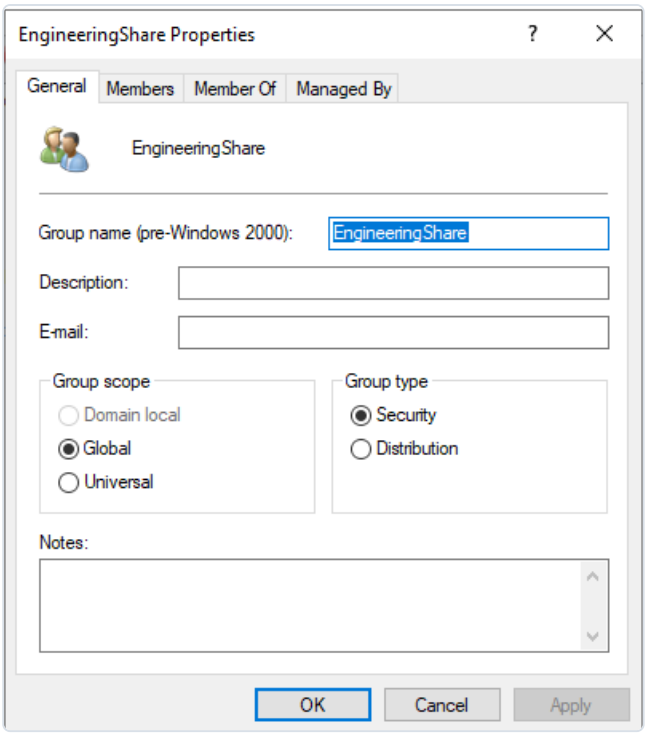


Figure 24 — Group scope Global, group type Security

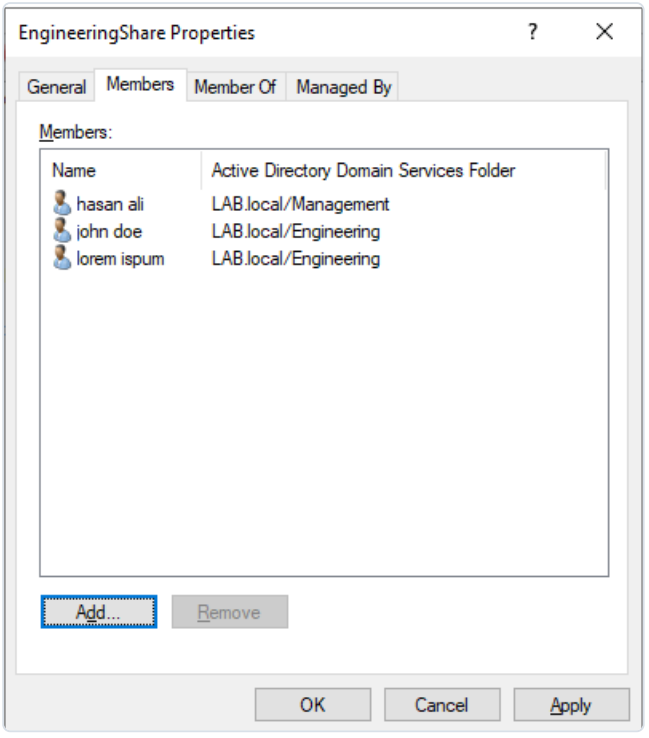


Figure 25 — Members: hasan ali (Management), john doe and lorem ispum (Engineering)

Part G — Protected file share (17 Oct)

G1. In *Server Manager → File and Storage Services → Shares → New Share*, created **SMB Share – Quick** at `C:\Shares\EngineeringShare` (network path `\\DC01\EngineeringShare`).

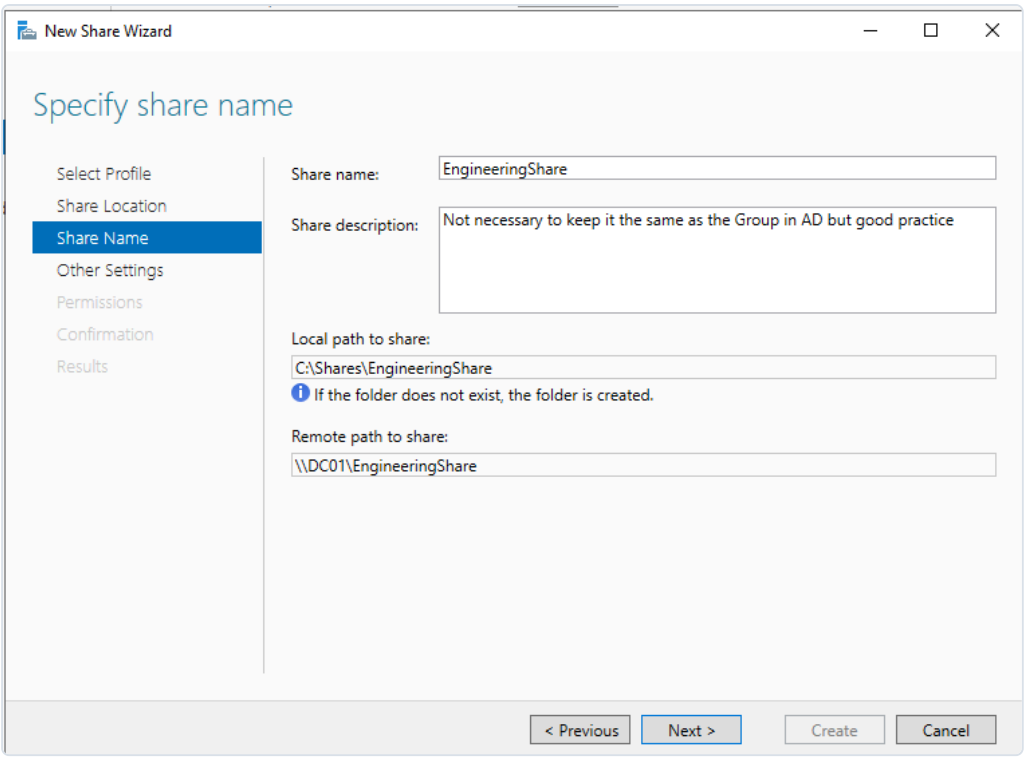


Figure 26 — Share name and paths

G2. Lock down the folder permissions (NTFS). I turned off inheritance (so the folder stops copying permissions from `C:\`), **removed the general "Users" group** (which would let every domain user in), and then gave the `EngineeringShare` group access.

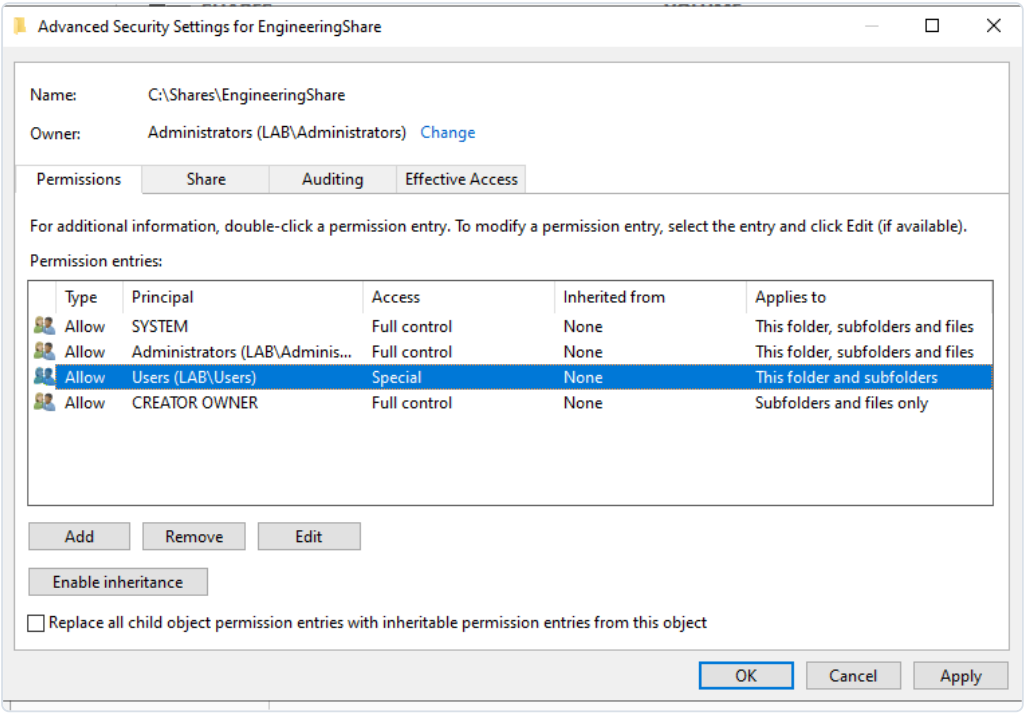


Figure 27 — Inheritance disabled; the general Users entry is still there

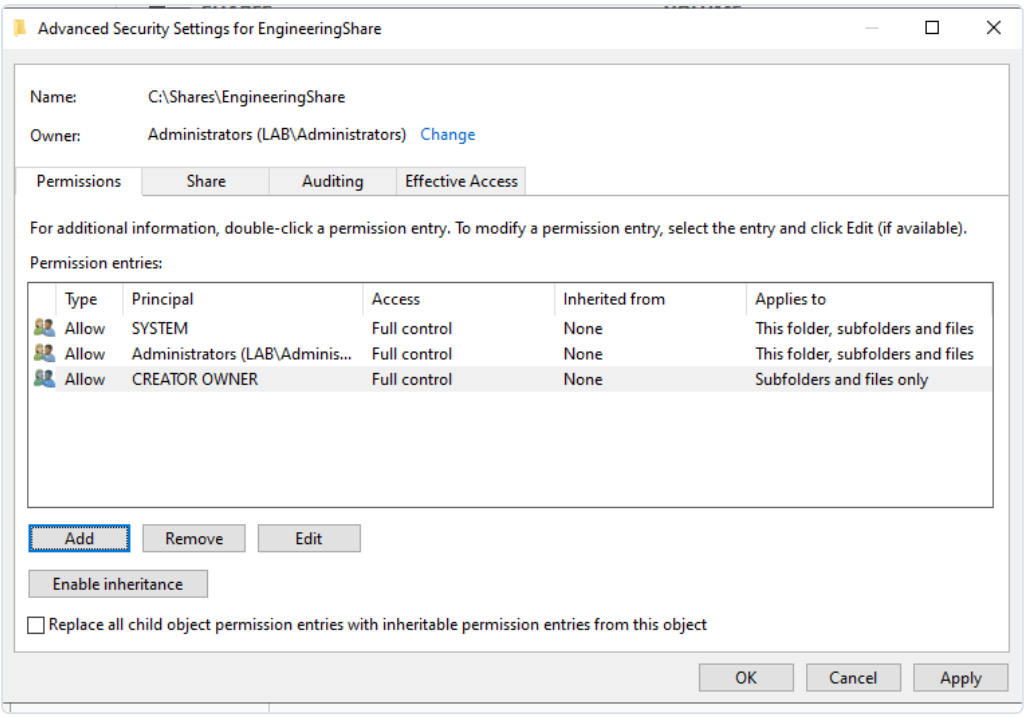


Figure 28 — Users group removed: only SYSTEM, Administrators and CREATOR OWNER left

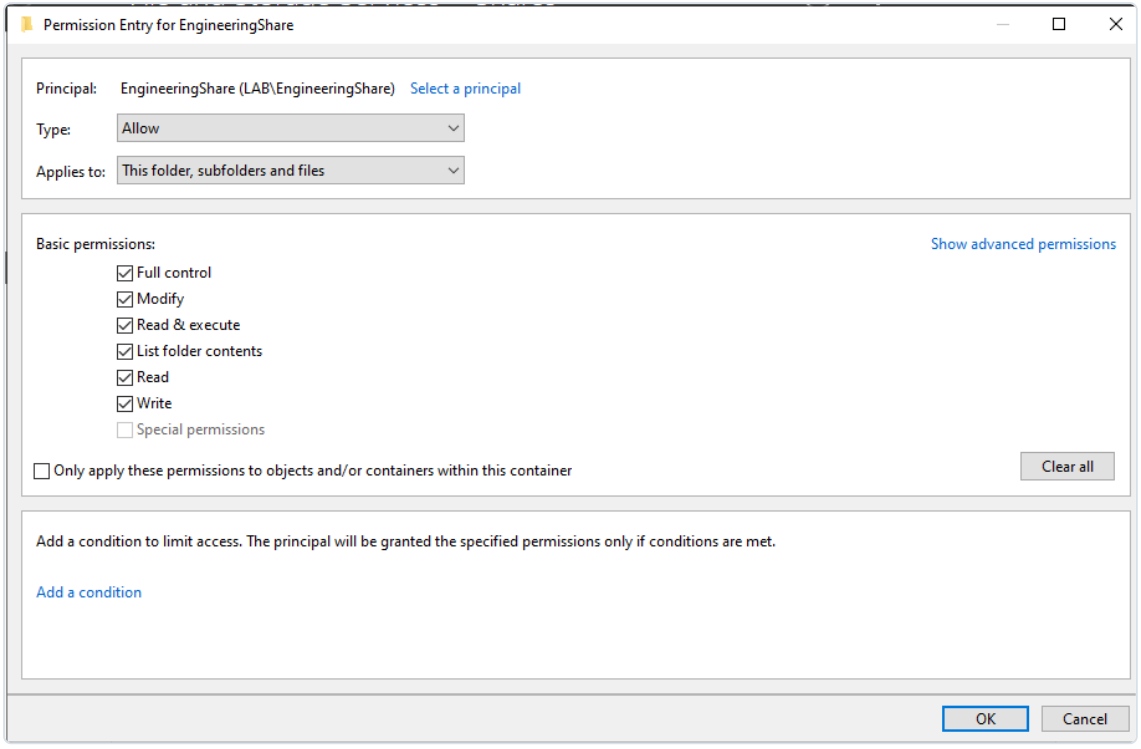


Figure 29 — EngineeringShare group added

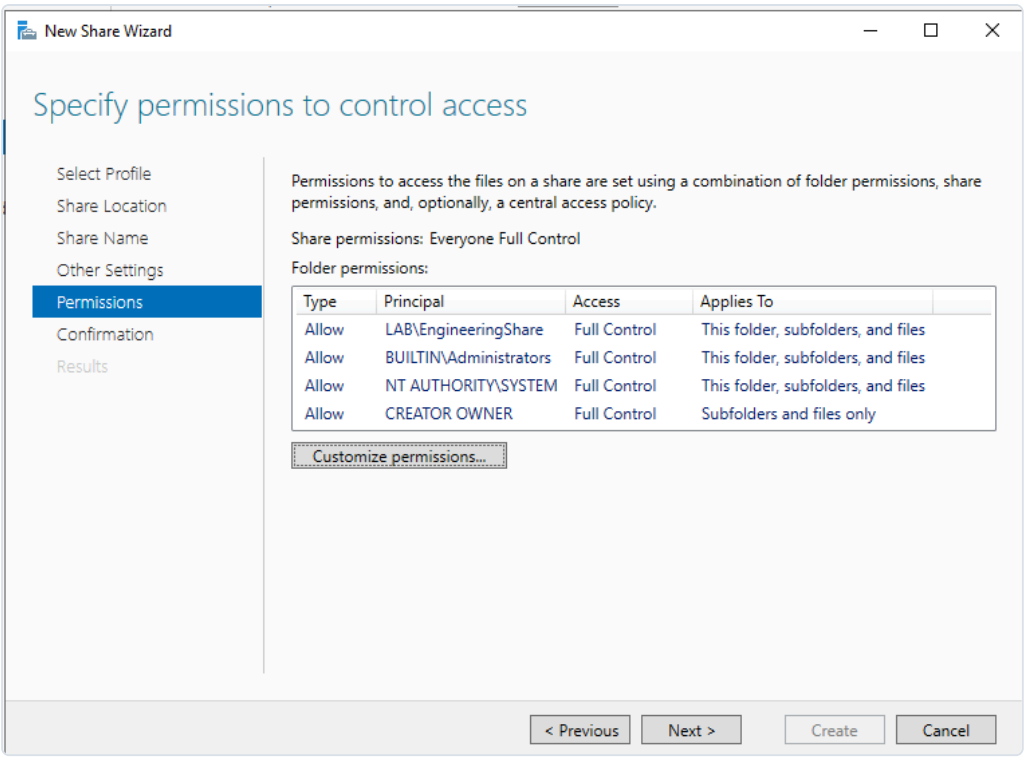


Figure 30 — Final permissions. Share level: Everyone. Folder (NTFS) level: only EngineeringShare, Administrators and SYSTEM.

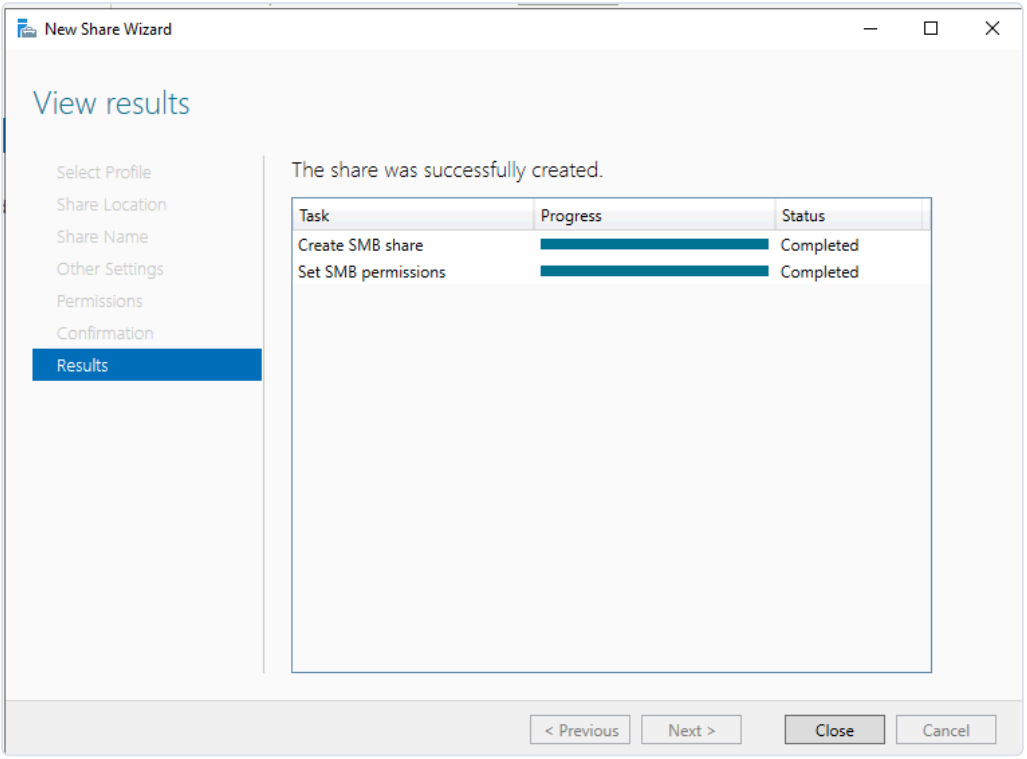


Figure 31 — Share created

Part H — Test who can open the share (17 Oct)

User	Department (OU)	In EngineeringShare group?	Expected	Result
Ahmed Rizwan	Management	No	Blocked	✅ Blocked
hasan ali	Management	**Yes**	Allowed	✅ Allowed — created a folder and mapped drive Z:

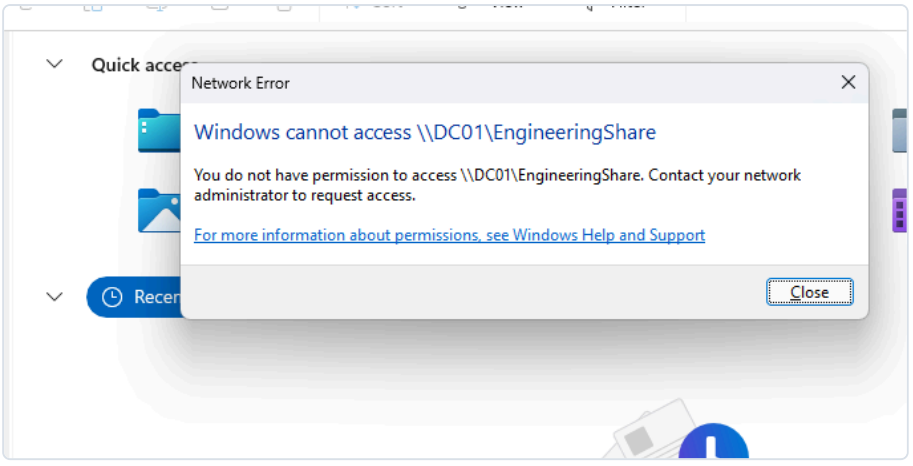


Figure 32 — Ahmed Rizwan: "You do not have permission to access \\DC01\EngineeringShare"

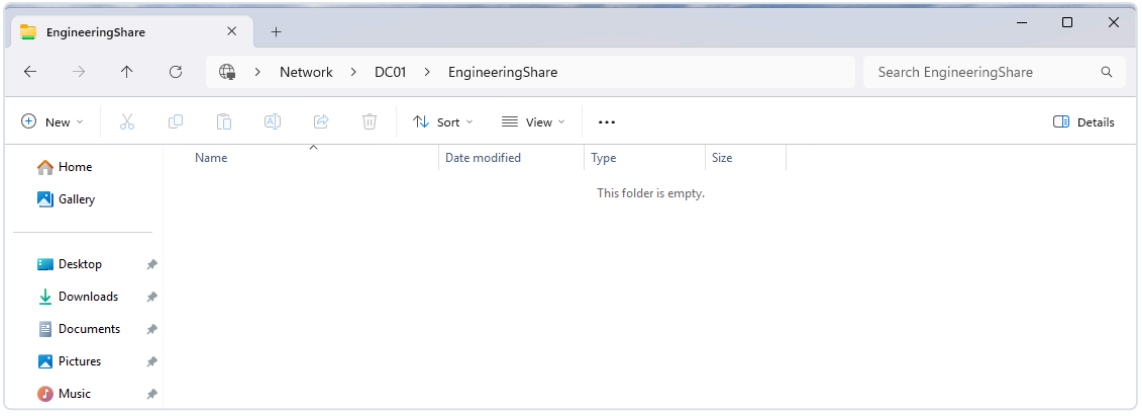


Figure 33 — hasan ali opens the share

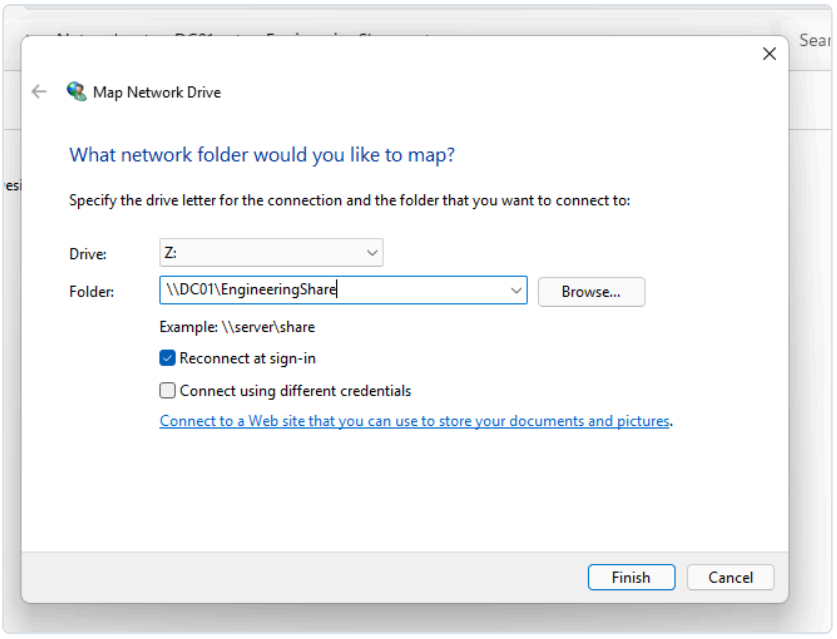


Figure 34 — Mapping the share as drive Z:, reconnect at sign-in

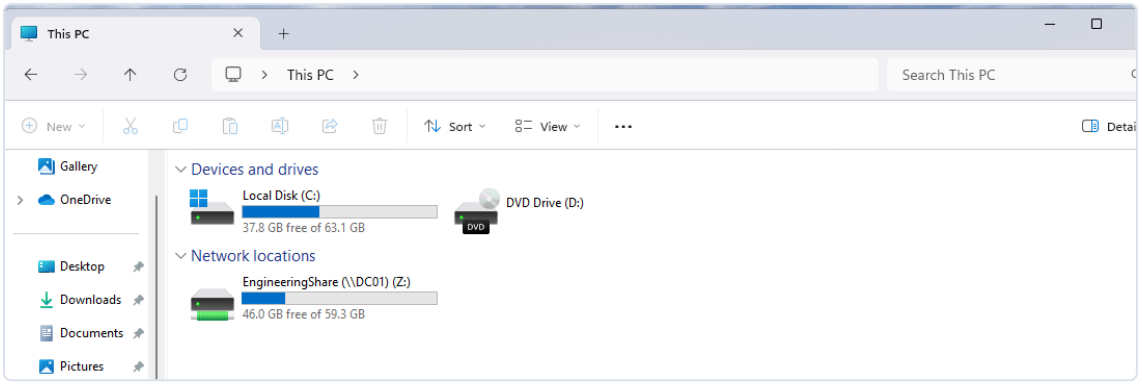


Figure 35 — EngineeringShare (\\DC01) (Z:) under Network locations

Part I — Group Policy: department wallpaper (17 Oct)

I1. Made an "Engineering Department" wallpaper and saved it in the server's **NETLOGON** share, a folder every domain computer can already read.

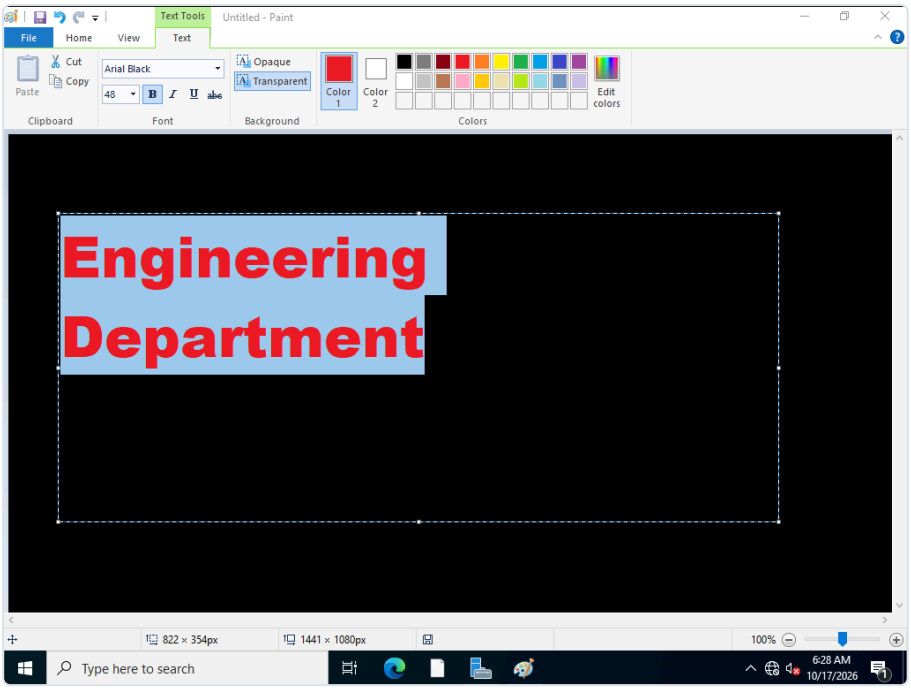


Figure 36 — Wallpaper image

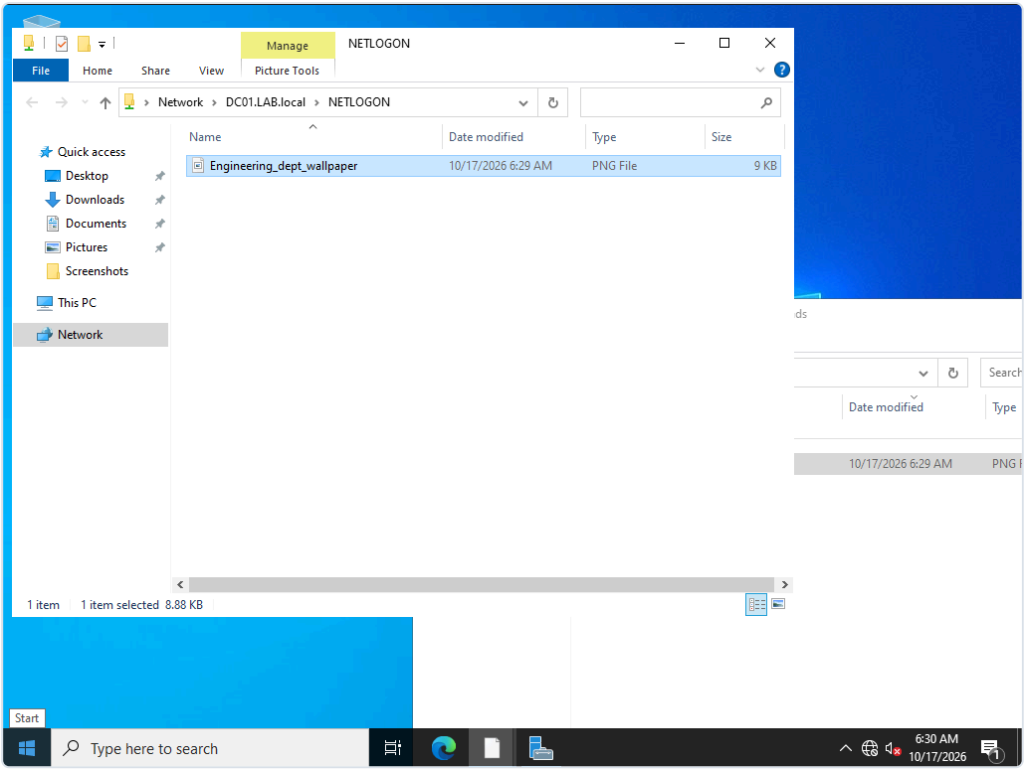


Figure 37 — Engineering_dept_wallpaper.png in \\DC01.LAB.local\NETLOGON

I2. In *Group Policy Management*, created and linked a GPO called `SetEngineeringWallpaper` to the **Engineering** OU, then set *User Configuration → Administrative Templates → Desktop → Desktop → Desktop Wallpaper* to the network path of the image.

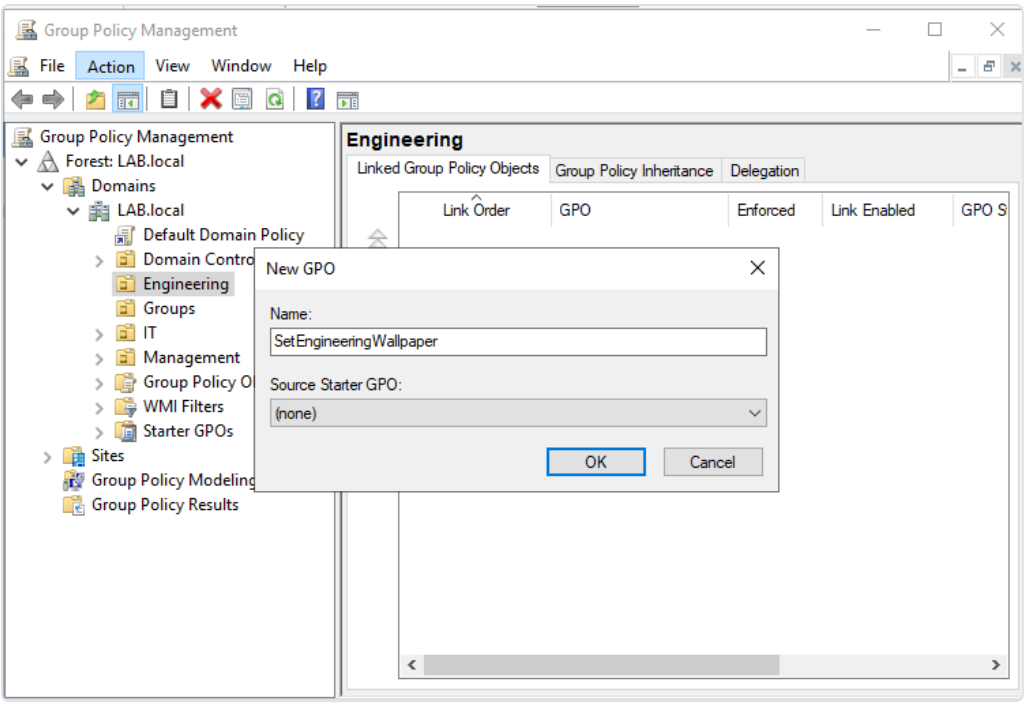


Figure 38 — Creating the GPO on the Engineering OU

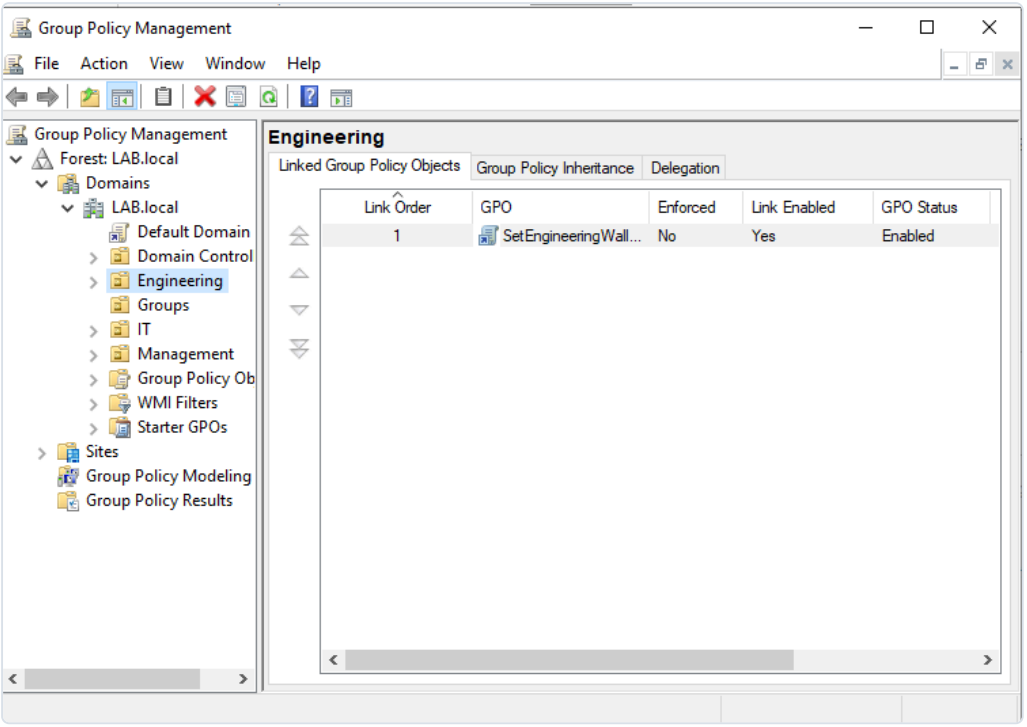


Figure 39 — Linked and enabled

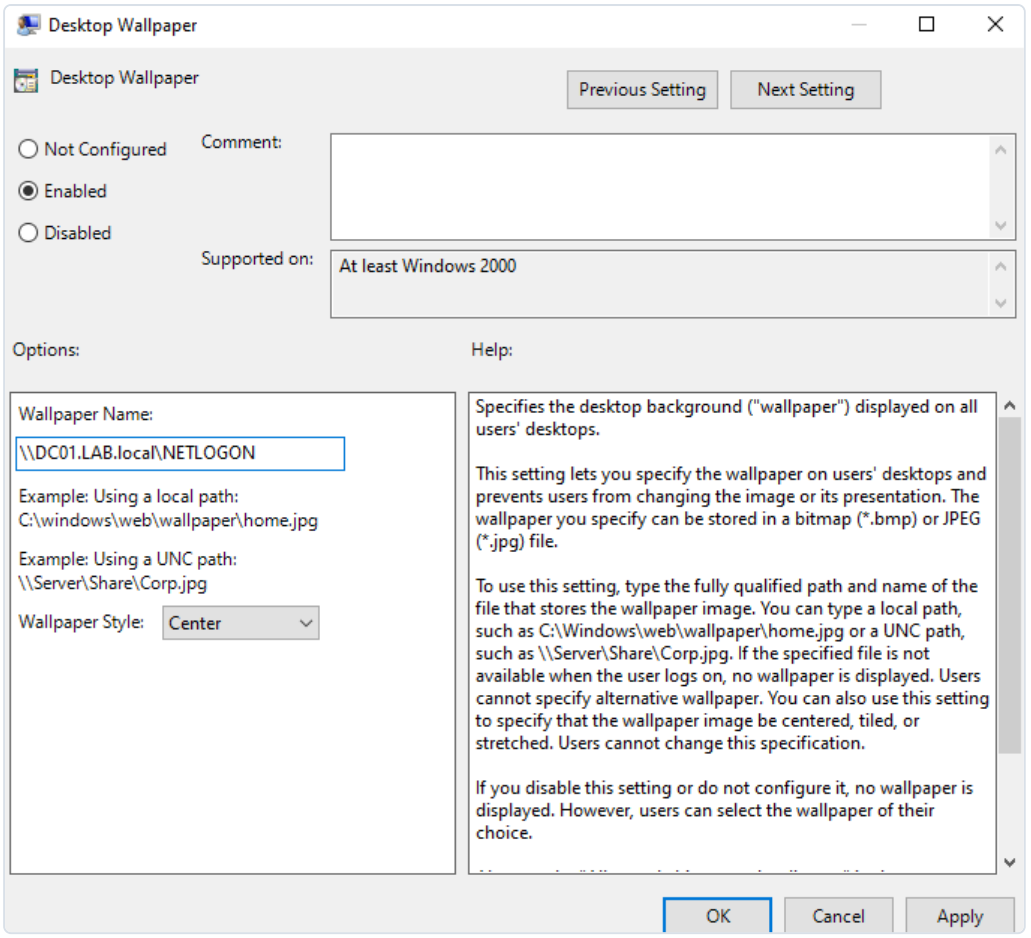


Figure 40 — Desktop Wallpaper enabled, pointing to \\DC01.LAB.local\NETLOGON\...

I3. Test: logged in as Engineering user **jdoe** — the wallpaper was applied automatically.

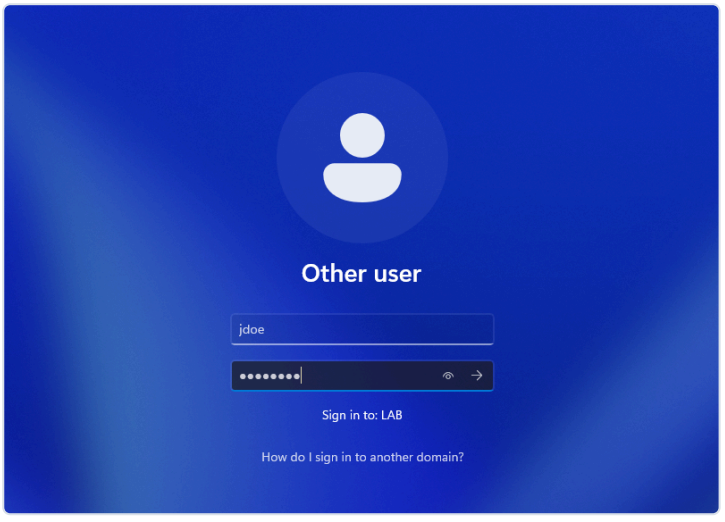


Figure 41 — Logging in as jdoe

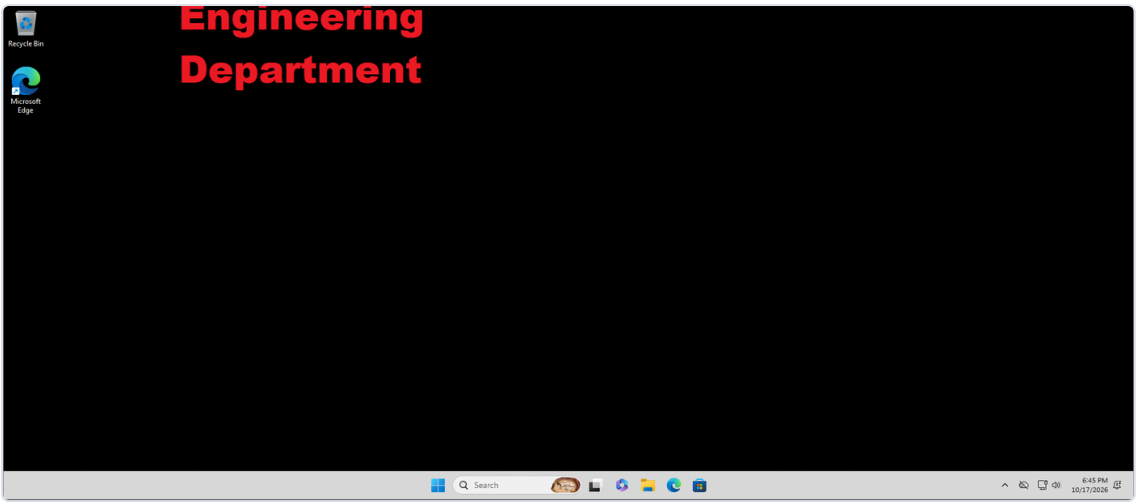


Figure 42 — Group Policy applied: Engineering wallpaper on the desktop

I4. Created a second GPO, `DisableChangingWallpaper`, and enabled *User Configuration → Administrative Templates → Control Panel → Personalization → Prevent changing desktop background*, so users cannot replace the company wallpaper.

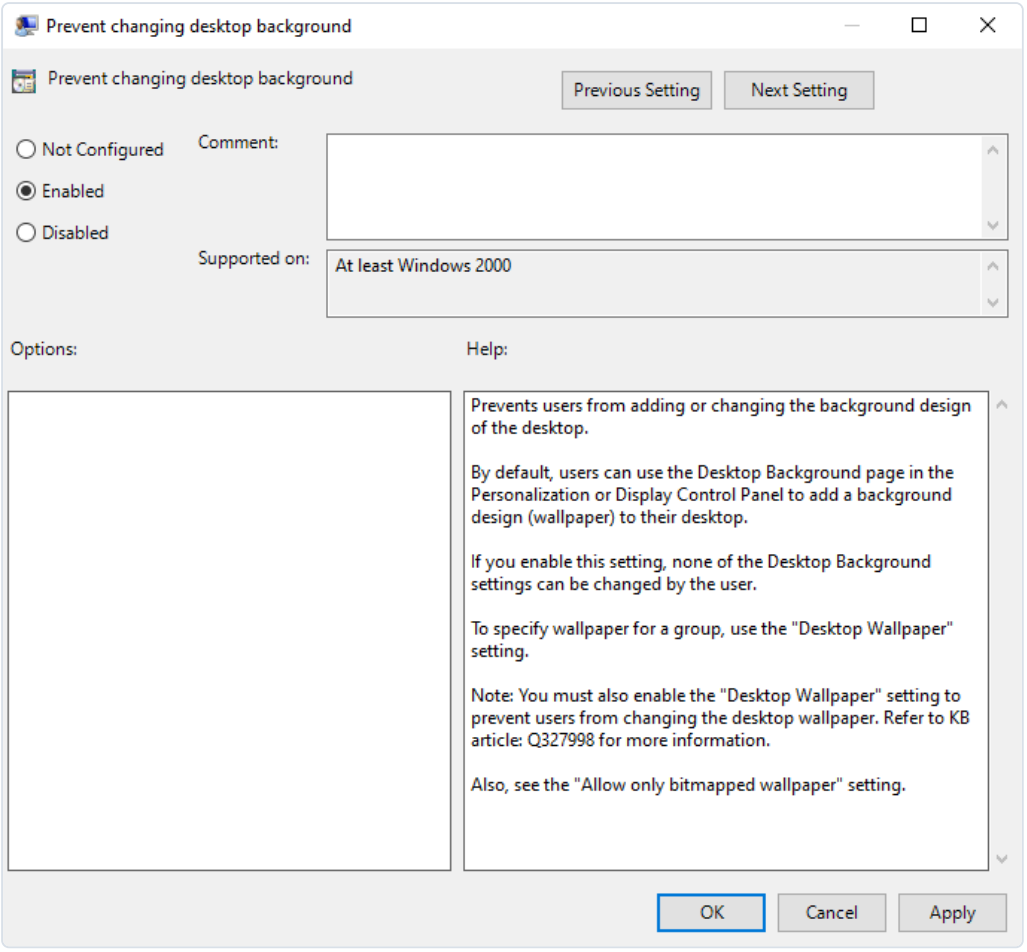


Figure 43 — "Prevent changing desktop background" enabled

5. PowerShell automation

`scripts/Create-ADUser.ps1` creates a domain user in one command instead of clicking through the wizard. It worked in the lab.

```
.\Create-ADUser.ps1 -FirstName John -Lastname Doe -Username jdoe -OU Engineering -Domain
LAB.local
```

What it does

1. Takes first name, last name, username, OU and domain as required inputs
2. Generates a random 12-character temporary password and shows it once
3. Converts the password to a secure string so it is handed to Active Directory safely
4. Creates the account enabled, in the chosen OU, with **"must change password at next logon"** turned on and password expiry left on

6. Test results

#	Test	Expected	Result
1	WS01 joins `LAB.local`	Join succeeds	✅ Pass
2	Domain user logs in on WS01	Login works	✅ Pass
3	Ahmed Rizwan (not in group) opens the share	Blocked	✅ Pass — blocked
4	hasan ali (in group) opens the share and creates a folder	Allowed	✅ Pass
5	Map the share as drive Z:	Drive appears	✅ Pass
6	Engineering user logs in	Engineering wallpaper applied	✅ Pass
7	`Create-ADUser.ps1` creates a user	User created	✅ Pass

7. Glossary of short forms

Short form	Meaning
AD	Active Directory — Microsoft's central list of users, computers and rules for a network
AD DS	Active Directory Domain Services — the server role that runs Active Directory
AD CS	Active Directory Certificate Services — the role that issues digital certificates
CA	Certificate Authority — the service that signs and issues certificates
DC	Domain Controller — the server that runs AD and checks every login
DNS	Domain Name System — turns names (like `DC01.LAB.local`) into IP addresses
DSRM	Directory Services Restore Mode — a recovery mode for a domain controller, with its own password
GPO	Group Policy Object — a set of rules pushed automatically to users or computers
IP	Internet Protocol address — a computer's number on the network
LDAPS	LDAP over SSL — encrypted directory lookups (LDAP = Lightweight Directory Access Protocol)
NAT	Network Address Translation — lets the VMs share the laptop's internet connection
NetBIOS	An older short name for the domain (`LAB`) used in logons like `LAB\ahrizwan`
NTFS	New Technology File System — Windows' disk format; its permissions control who can open a folder
OU	Organisational Unit — a folder inside AD for grouping users or computers (e.g. a department)
SMB	Server Message Block — the protocol Windows uses for shared folders
UNC path	Universal Naming Convention — a network path like `\\DC01\EngineeringShare`
VM	Virtual Machine — a computer running inside software on my laptop

Disclaimer

This is a learning lab built for practice. I will further tighten the security controls on **DC01** and **WS01** as the project continues.

Repository layout

```
AD-Homelab/
├── README.md                 ← this page
├── AD-Homelab-Report.pdf     ← PDF copy of this page
├── images/                   ← screenshots used above
└── scripts/
    └── Create-ADUser.ps1
```